Consent-Driven Smart Attendance System Using Facial Recognition and LMS Synchronization

Team

JetSet - Group 3

Team Lead

Sorel Agbogla

Team Members

Sorel Agbogla
Andres Moreira
Andrew Kim
Shayan Karki

Sponsors

Diana Rabah
Jordan Black

3/1/2026

1. Introduction

This document presents the Phase 2 design specification for the Consent-Driven Smart Attendance System Using Facial Recognition and LMS Synchronization. The purpose of this phase is to define the system architecture, database structure, behavioral workflows, and data processing logic prior to implementation. The diagrams included in this document provide a comprehensive blueprint to ensure feasibility, scalability, and alignment with stakeholder requirements.

2. System Design Architecture Diagram

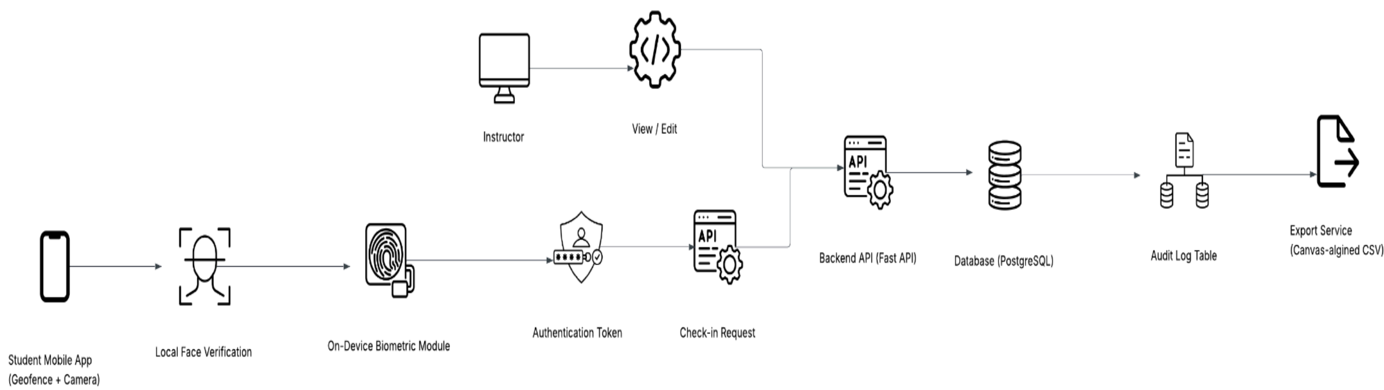

Figure 1: System Architecture Diagram

The system architecture illustrates the interaction between the student mobile application, instructor interface backend application layer, database layer, geofence validation, and biometric verification processes. The backend manages attendance sessions, records attendance status, and applies manual overrides when necessary, The layered structure supports modular design, secure data handling, and integration with external Learning Management Systems (LMS).

3. Data Flow Diagram (DFD)

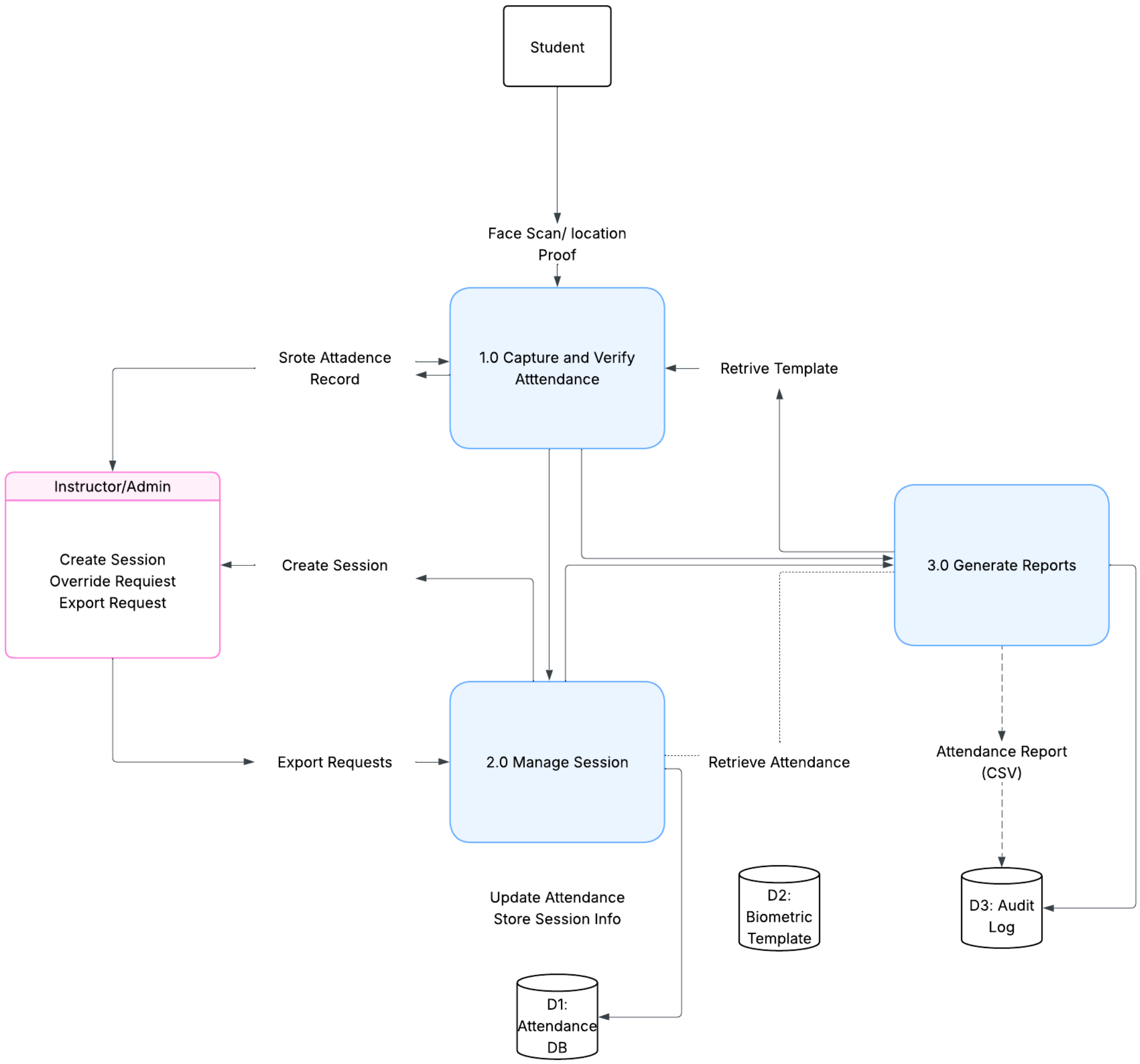

Figure 2: Data Flow Diagram

The data Flow Diagram represents how attendance data moves through the system. Student biometric verification results and geolocation data are submitted during an active attendance session. The system validates geofence boundaries and records biometric verification outcomes within the ***AttendanceRecord*** entity. Instructor actions such as session creation, attendance view, and manual overrides are processed and stored in the database for auditing and reporting purposes.

4. Entity Relationship Diagram (ERD)

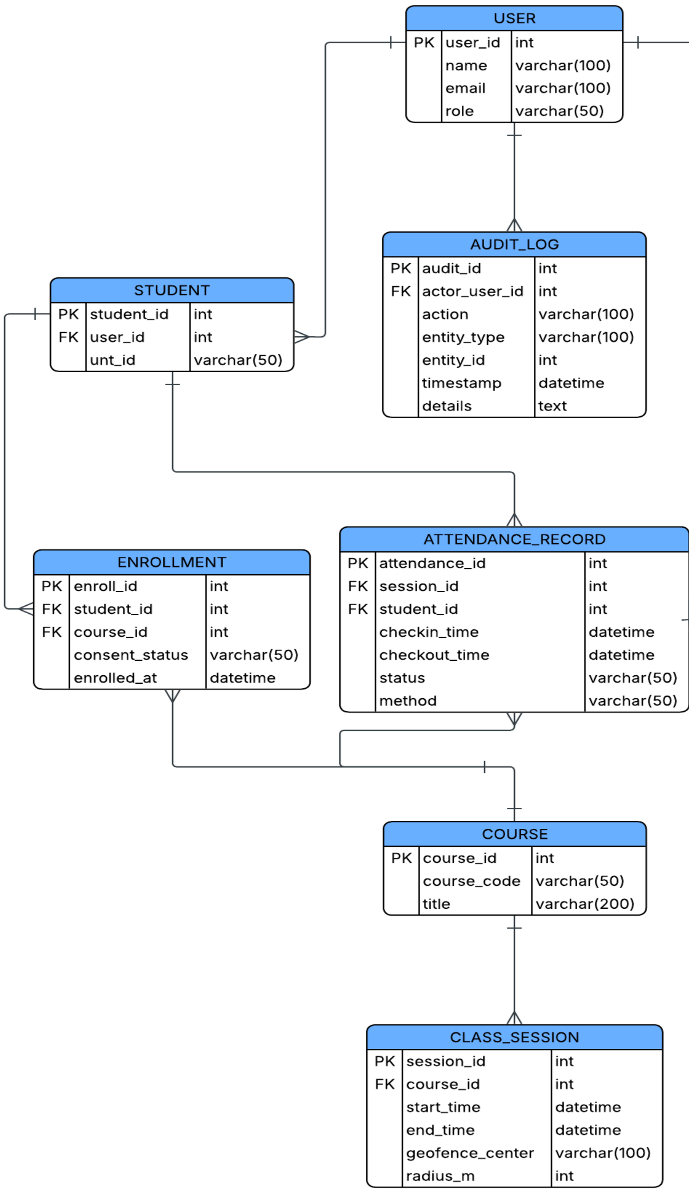

Figure 3: Entity Relationship Diagram

The ERD defines the relational database structure supporting the system. It models core entities including ***Users***, ***Students***, ***Instructors***, ***Courses***, ***AttendanceSessions, AttendanceRecords, Classrooms***, ***Geofences***, ***BiometricData, ManualOverrides***. Primary and foreign key relationships ensure referential integrity between sessions, records, and associated users while supporting accurate attendance tracking and LMS synchronization.

5. UML Class Diagram

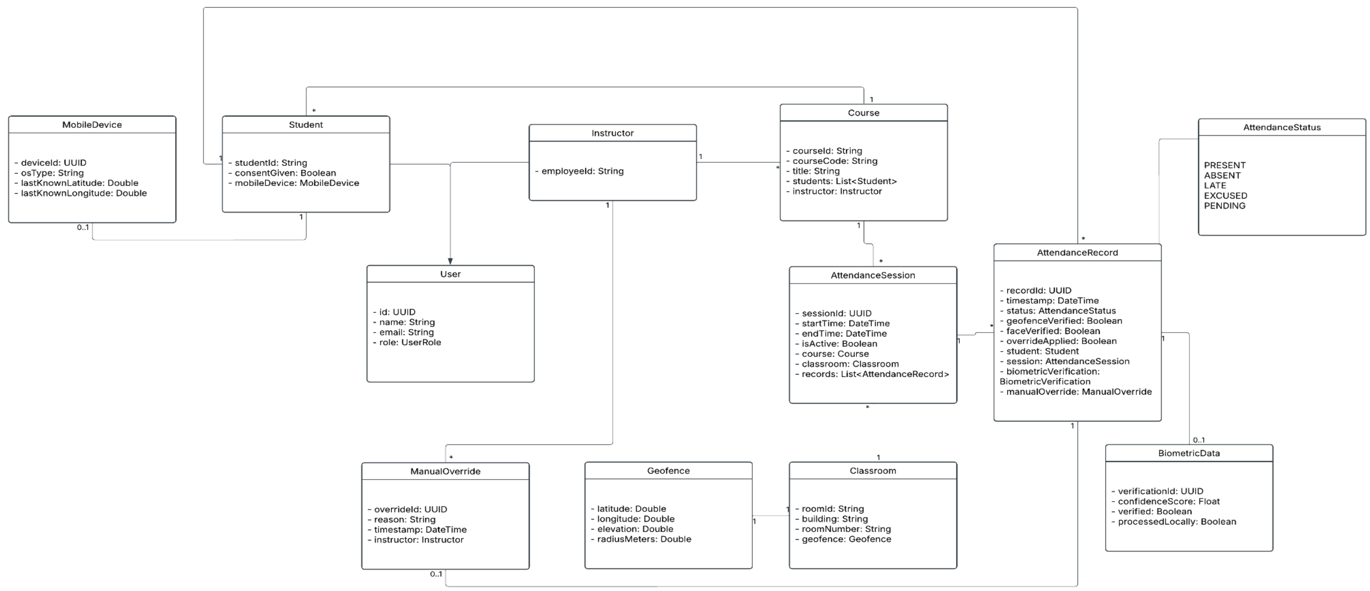

Figure 4: Class Diagram

The UML Class Diagram models the object oriented structure of the system. It defines inheritance relationships between ***User***, ***Student***, and ***Instructor classes***, as well as associations between Course, ***AttendanceSession,*** and ***AttendanceRecord*** entities. Additional classes such as ***BiometricData, Geofence, Classroom,*** and ***ManualOverride*** represent supporting components required for attendance validation and state tracking.

6. State Transition Diagram (Student Attendance Flow)

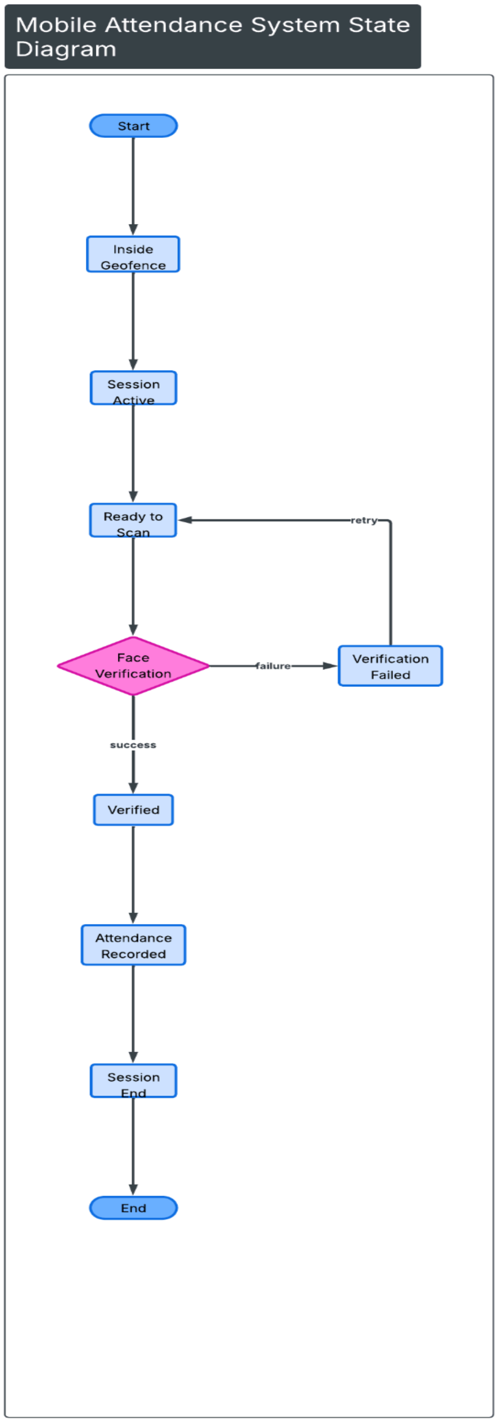

Figure 5: State Transition Diagram

The state Transition Diagram models the dynamic behavior of student attendance within a session. It illustrates the progression from geofence validation to biometric verification, attendance confirmation, and potential instructor override. State changes reflect system conditions such as verification success, session closure, or manual status adjustment, ensuring clear workflow logic and controlled transitions.